SURAKSHIT KESSOONATH

COURSE: BSc (Hons) COMPUTER SCIENCE WITH NETWORK SECURITY STUDENT

CONTACT

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PERSONAL DETAILS

Nationality: Mauritian
Driving Licence: Yes

LANGUAGES

English — Fluent
French — Fluent
Hindi — Fluent
Mauritian Creole - Native

HOBBIES & INTERESTS

- Football
- Gym
- Reading & Music

REFEREES

References available on request

PROFILE

Second-year BSc Computer Science with Network Security student at the University of Technology, Mauritius, with a background in Mathematics, Accounting and Economics. Focused on where software development and cybersecurity intersect, with hands-on project experience across full-stack development, networking and applied cryptography.

EDUCATION

University of Technology, Mauritius - BSc Computer Science with Network Security Apr 2025 – Present

Year 1 completed; now Year 2, Sem 1: Programming Techniques 2, Data Security & Cryptography, OS Administration, Server-Side Web Programming, Switching & Routing Techniques.

Cambridge International A-Levels Jan 2017 - Nov 2024
Sir Abdool Razack Mohamed State Secondary School — Mathematics, Accounting, Economics; SC & HSC Pass.

Modules &Technical Projects

Aqua-Lux — Secure Aquarium E-Commerce Platform | Micronaut, Java 21, MySQL 8, JavaScript, Tomcat, Docker Compose

- Built a three-tier REST API application with token-based session authentication, SQL-injection prevention and password hashing, fully containerized with Docker Compose and documented (API, architecture, data model).

Gym-Pro Manager — Gym Management System | Java Swing, JDBC, PostgreSQL, D-Beaver

- Developed an object-oriented desktop application using inheritance, interfaces and polymorphism, with multi-step JDBC commit/rollback transactions and ~35 documented test cases covering input validation, SQL-injection resistance and database-level verification.

Enterprise Network Design & Simulation | Cisco Packet Tracer, Cisco IOS

- Designed and configured 2 routers, 3 switches (incl. a Layer 3 switch) and 4 VLANs, implementing router-on-a-stick inter-VLAN routing, 802.1Q trunking, PAgP EtherChannel, OSPF, SVI management addressing and SSH, with verified end-to-end connectivity.

Network Defence & Forensics Lab | Cisco IOS, Nmap/Zenmap, Wireshark, Ubuntu

- Configured standard and extended ACLs to filter hosts, ICMP and HTTP traffic, performed port/service scanning with Nmap and live packet capture with Wireshark, and ran SHA-256 file-integrity monitoring with Linux permission correction.

Applied Cryptanalysis Coursework | Classical & symmetric cryptography, Python

- Broke substitution, Vigenère, columnar transposition and XOR key-reuse ciphertexts with documented key-recovery reasoning, and analyzed an S-box cipher variant and symmetric key distribution/management.

Road Traffic Accident Analysis | Python, pandas, Matplotlib/Seaborn, scikit-learn, Jupyter

- Cleaned and explored a real accident dataset, visualized temporal and conditional patterns, and trained/evaluated a classification model predicting accident severity with justified algorithm and metric choices.

Arduino Circuits & Autonomous EV3 Robot | Arduino, C/C++, Lego EV3, sensors & actuators

- Built LED control interfaces, a digital timer and a mini electronic keyboard on Arduino, and programmed a Lego EV3 robot for sensor-based navigation through iterative testing and a final obstacle-course race.

Other coursework: Oracle database design (6–10 entity schema, constraints, PL/SQL, users & privileges); CPU fetch–decode–execute simulator; FCFS/SJF/Round-Robin schedulers; FIFO/LRU/Optimal page-replacement simulations (Python).

PROFESSIONAL COURSES

Robotics & Embedded Systems — Arduino & EV3 Robot (July 2026)

- Built LED, timer, and keyboard circuits using Arduino
- Programmed a Lego EV3 robot for sensor-based navigation
- Networking Basics (Cisco Networking Academy) (May 2026)
- Ethical Hacker (Cisco Networking Academy) (May 2026)

SKILLS

Networking & Security: VLANs, routing (OSPF/static), ACLs, EtherChannel, SSH, ethical hacking basics
Security Tools & Cryptography: Nmap, Wireshark, applied cryptography
Programming: Java, JavaScript, Python (pandas, scikit-learn), REST APIs, JDBC, Java Swing
Databases: SQL & PL/SQL, MySQL, PostgreSQL, Oracle
Systems & Cloud: Linux administration, Bash, Docker & Docker Compose, AWS fundamentals, Git & GitHub
Embedded & Other: Arduino, microcontrollers, circuits, technical documentation, teamwork, problem-solving

ACHIEVEMENTS & EXPERIENCE

- Event Volunteer, School Functions & Sports Day (2024) •
- Ceramic & Pottery Workshop, Rabindranath Tagore Institute (2023) •
- Internet & Computing Core Certificate – IC3 (2019) •
- Sports Day — 2 Gold, 1 Bronze Medal (2018) •
- Student Council Member (2017)